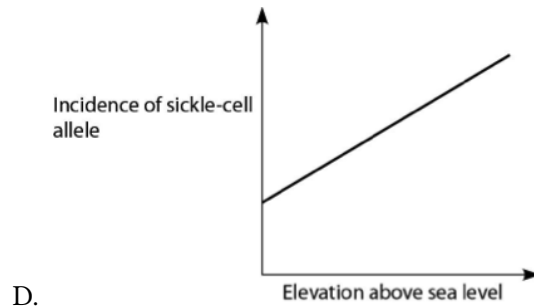
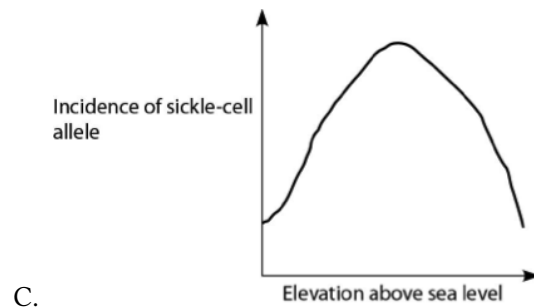
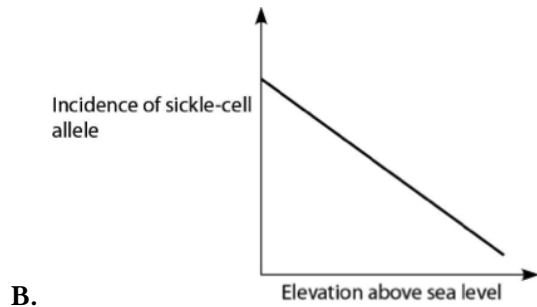
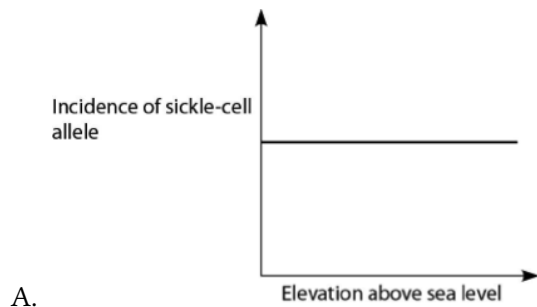


Chill. It's just a few pieces of paper. Each question is worth 2 points except where noted.

Multiple choice. Choose the *best* answer. Clearly circle the letter for your answer. Vague or unclear marks will be counted as incorrect.

1. Which *one* of the following statements is true of natural selection?
 - A. It involves differences in reproductive success.
 - B. It requires genetic variation, results in descent with modification, and involves differences in reproductive success.**
 - C. It requires genetic variation.
 - D. It results in descent with modification and involves differences in reproductive success.
 - E. It results in descent with modification.
 2. Which mouse has the greatest relative fitness? (A clutch is the offspring from one breeding event.)
 - A. One that lives for 4 years, has 3 clutches per year, and 4 offspring per clutch.
 - B. One that lives for 2 years, has 3 clutches per year, and 8 offspring per clutch.
 - C. One that lives for 3 years, has 3 clutches per year, and 5 offspring per clutch.
 - D. One that lives for 3 years, has 3 clutches per year, and 7 offspring per clutch.**
 - E. One that lives for 3 years, has 2 clutches per year, and 6 offspring per clutch.
 3. The overuse of drugs such as penicillin has caused an increase in the number of bacterial species that are resistant to the drugs. The most likely explanation for this result is:
 - A. The drugs caused mutations in the bacteria that gave them resistance.
 - B. Most bacteria can resist drugs.
 - C. Bacteria began making proteins resistant to the drugs.
 - D. A few bacteria were naturally resistant and natural selection favored their increase.**
 - E. Penicillin was not an effective drug.
 4. According to Darwin's concept of descent with modification:
 - A. All species have evolved from a single common ancestor.
 - B. Species evolve from ancestral forms through the accumulation of different adaptations.**
 - C. Populations evolve by natural selection acting on individuals.
 - D. All organisms overproduce offspring, leading to different adaptations.
 - E. All organisms overproduce offspring, leading to differences in survivorship.
 - F. Individuals with greater fitness will evolve more quickly than individuals with lower fitness.
 5. Which of these conditions should completely prevent the occurrence of natural selection in a population over time?
 - A. All variation between individuals is due only to environmental factors.**
 - B. The environment is changing at a relatively slow rate.
 - C. The population mates non-randomly.
 - D. The population size is large.
 - E. The population lives in a habitat where there are no competing species present.
-

6. *Anopheles* mosquitoes, which carry the malaria parasite, cannot live above elevations of 5,900 feet. In addition, oxygen availability decreases with higher altitude. Consider a hypothetical human population that is adapted to life on the slopes of Mt. Kilimanjaro in Tanzania, a country in equatorial Africa. Mt. Kilimanjaro's base is about 2,600 feet above sea level and its peak is 19,341 feet above sea level. If the frequency of the sickle-cell allele in the population is plotted against altitude (feet above sea level), which of the following distributions is most likely, assuming little migration of people up or down the mountain?



7. The rabbitsfoot mussel was once widespread throughout rivers of eastern North America but now remains in only a few very small populations. The decrease occurred rapidly due to habitat loss. Genetics studies show that the remaining populations have little or no genetic variation. The best explanation for this result is:
- Balancing selection favored only one or two alleles.
 - A bottleneck caused the loss of homozygosity.
 - Directional selection favored only one or two alleles.
 - Inbreeding caused an increase in heterozygosity.
 - Genetic drift caused the loss of heterozygosity.**
8. Organisms living in the northern rocky intertidal face competing selective forces. Larger individuals benefit by having a larger internal volume, which helps to conserve body temperature. However, smaller individuals are less likely to be dislodged by large waves from storms. Thus, selection may favor intermediate body size phenotype for any particular species. The mode of selection is
- natural selection.
 - disruptive selection.
 - stabilizing selection.**
 - phenotype selection.
 - directional selection.

9. Violation of these two Hardy-Weinberg assumptions always causes homozygosity to increase.
- A. Genetic drift and natural selection.
 - B. Non-random mating and gene flow.
 - C. Gene flow and mutation.
 - D. Genetic drift and non-random mating.**
 - E. Natural selection and gene flow.
 - F. Mutation and natural selection.
10. In science, a hypothesis is only useful if it
- A. agrees with the conclusion.
 - B. explains the natural world.
 - C. is proven correct.
 - D. can be proven correct.
 - E. can be tested.**
11. (5 points) Here is the full classification of the Bluntnose Sixgill Shark.
- Eukaryota; Metazoa; Chordata; Chondrichthys; Hexanchiformes; Hexanchidae; Hexanchus; griseus
- (a) Which is the family? **Hexanchidae**
- (b) Write the scientific name using proper binomial nomenclature. ***Hexanchus griseus***

Circle True or False. Read each statement completely before answering.

12. **TRUE** False Adaptations can lead to descent with modification.
13. True **FALSE** The genotype is determined by the interaction of the phenotype with the environment.
14. True **FALSE** One assumption of the Hardy-Weinberg equations is that mutations are random.
15. **TRUE** False Genetic drift always increases homozygosity in a population.
16. **TRUE** False The smallest unit of evolution is the population.
17. True **FALSE** Overproduction of offspring by every individual in a population is necessary for evolution to occur.
18. **TRUE** False If $p = 0.4$, then total homozygosity in the population is 0.52 (assume equilibrium).
19. **TRUE** False Sexual selection usually causes a pattern of directional selection.
20. **TRUE** False Genetic drift is strongest in small populations.
21. True **FALSE** Thomas Robert Malthus wrote *Systema Naturæ* that influenced Darwin.
-

Match the *most appropriate* term to each concept. Not all terms are used. No term is used twice.

adaptation	environment	Thomas Robert Malthus
allelotype	genotype	mutation
artificial selection	haploid	natural selection
balancing selection	independent evolution	phenotype
convergent evolution	inductive	population selection
Georges Cuvier	Jean Baptiste de Lamarck	scientific
deductive	Carolus Linnaeus	stabilizing selection
diploid	Charles Lyell	Alfred Russel Wallace

22. **Adaptation** Trait that increases fitness in the current environment.
23. **Environment** Contributes to the phenotype but is not inherited.
24. **Balancing Selection** Can cause stabilizing selection in a population.
25. **Diploid** Has two alleles at a gene.
26. **Convergent evolution** Unrelated organisms evolve similar traits by adapting to similar habitats.
27. **Inductive** Reasoning that uses specific observations to develop general principles.
28. **Mutation** Creates new genetic variation within a population.
29. **Genotype** The combination of alleles at a gene.
30. **Lamarck** Was the first to propose that organisms change over time.
31. **Lyell** Argued that Earth changed uniformly for much more than 6000 years.
32. Briefly compare sexual selection to natural selection to explain why sexual selection can cause evolutionary change more rapidly than other forms of natural selection.

Sexual selection acts directly on reproductive success while natural selection is indirect.

You may use a calculator to solve the following problem but you must **show your work** in the space provided. You may **round to 3 decimal places after each step or after the final calculation**. Any other rounding will result in a 3 point deduction.

33. (15 points) Impatiens is a type of flower that has three flower colors. Red color is produced by RR individuals. Pink color is produced by Rr individuals. White color is produced by rr individuals. A random sample of 248 flowers found that 42 were red, 175 were pink, and 31 were white. Calculate the two allele frequencies and the three genotype frequencies. *Show whether the population is in Hardy-Weinberg equilibrium.*

The final values may sum to 0.999, 1.000 or 1.001 due to rounding error. (2 pts each plus 5 points for showing your work.)

Frequency	
R	0.522
r	0.478
RR	0.169
Rr	0.706
rr	0.125

Be sure to show whether the population is in Hardy-Weinberg equilibrium and *show all work*.

Sums to 1.0.

34. (16 points) List below the four missing steps of the Scientific Method. Then, for each step, write a statement that illustrates how scientists applied the scientific method to the experiments described below. I have provided step four as an example.

For centuries, people noticed that plants tend to grow or lean towards a window but, prior to the 18th century, no one knew why plants responded this way. During the late 1600s to early 1700s, scientists considered several possible explanations. One idea was that plants responded to fresh air. An experiment was designed to grow several plants in two growth chambers, with light coming in from the side by a window. The only difference between the chambers was that one had the window closed so light could enter the chamber but fresh air could not enter. The other chamber had the window open so that light and fresh air could enter. The plants in both chambers grew towards the window. The results of the experiment showed that fresh air was not the reason the plants were growing or leaning towards windows.

(a) **Observation:** Plants leaned towards the window.

(b) **Hypothesis:** Plants are responding to fresh air from the window.

(c) **Prediction:** If the window is open, then plants will lean toward the window.

(d) **Experiment and results:** Scientists noted that plants leaned towards a light source but not to a fresh air source or a warm temperature source.

(e) **Conclusion:** The hypothesis was falsified.

2 points for each step and 2 points for each explanation.
